ALICE L WALTON SCHOOL OF MEDICINE

Hypotension: Causes, Consequences, and Management

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Learning Objectives (4)

After completing this brick, you will be able to:

1. Define hypotension and identify its primary hemodynamic effects on organ perfusion and tissue oxygenation.
2. Describe the pathophysiological mechanisms leading to hypotension.
3. Identify risk factors for hypotension.
4. Outline nonpharmacological and pharmacological management strategies for hypotension.

CASE CONNECTION

EL, a 56-year-old female, presents to the emergency department with dizziness, fatigue, and recent episodes of fainting. During internal medicine rounds, medical student TR learns that EL's blood pressure is markedly low at 85/50 mmHg. She reports feeling lightheaded when standing up, and her symptoms have worsened over the past week. Her medication list includes hydrochlorothiazide and lisinopril for hypertension.

What could be causing EL's hypotension? What are the likely risks for a patient like her? Consider your answers as you read, and we'll revisit EL at the end of the brick.

GO TO CONCLUSION ↓

What Is Hypotension and What Are Its General Hemodynamic Effects? (LO#1)

Hypotension refers to abnormally low blood pressure that can compromise organ perfusion and tissue oxygenation. While blood pressure naturally fluctuates throughout the day in response to various physiological stimuli, sustained low blood pressure can lead to inadequate blood flow to vital organs, potentially resulting in organ dysfunction or failure.

Definition and Criteria

Hypotension is generally defined as a systolic blood pressure (SBP) below 90 mmHg or a mean arterial pressure (MAP) less than 65 mmHg. While these thresholds are commonly used in clinical practice, it's important to note that the definition can vary based on the clinical context and the individual patient's baseline blood pressure. For instance, a patient with chronic hypertension may experience hypotensive symptoms at blood pressure levels that would be considered normal in other individuals due to their body's adaptation to higher baseline pressures.

Several classification systems exist for hypotension based on etiology, clinical presentation, and hemodynamic parameters. Orthostatic hypotension is defined as a drop in systolic blood pressure of at least 20 mmHg or diastolic blood pressure of at least 10 mmHg within three minutes of standing. Postprandial hypotension refers to a decrease in systolic blood pressure of at least 20 mmHg within two hours after eating.

What is orthostatic hypotension?

Hemodynamic Changes

Hypotension is a decrease in blood pressure (BP), defined by **BP** = **cardiac output** × **systemic vascular resistance**. This is a variation of CO = ΔP/R. Compensatory cardiovascular responses—tachycardia, increased contractility, and vasoconstriction—aim to preserve perfusion to vital organs, particularly the brain and heart. Sympathetic and hormonal mechanisms reduce blood flow to skin, muscle, and splanchnic tissues. Severe or prolonged hypotension causes inadequate tissue perfusion, cellular hypoxia, and anaerobic metabolism with lactic acidosis. Persistent ischemia triggers inflammation and oxidative stress, leading to multi-organ dysfunction.

What defines hypotension and what is its primary hemodynamic effect?

What Is the Pat e the Common Causes of Hypotension? (LO#2)

Hypotension develops when the physiological, homeostatic mechanisms that regulate blood pressure are disrupted. These disruptions can occur through various pathways, affecting either cardiac output, systemic vascular resistance, or intravascular volume—the three primary determinants of blood pressure. The causes of hypotension can be broadly classified into three major categories: impaired vascular tone, reduced cardiac output, and volume depletion.

Impaired Vascular Tone

Autonomic neuropathy, often seen in diabetes or amyloidosis, can impair the body's ability to maintain vascular tone through sympathetic stimulation of the heart and vessels. Medications represent another significant cause of vasodilation, with antihypertensives (particularly calcium channel blockers, alpha-blockers, and nitrates), anesthetics, and antipsychotics being common culprits.

Endocrine disorders such as adrenal insufficiency (Addison's disease) and hypothyroidism can also impair vascular tone due to deficiencies in hormones that help maintain blood pressure.

Systemic inflammatory responses can profoundly affect vascular tone through the release of inflammatory mediators such as cytokines, nitric oxide, and prostaglandins. This mechanism is central to septic shock, anaphylaxis, and **systemic inflammatory response syndrome (SIRS)** causing excessive vasodilation.

Reduced Cardiac O

Reductions in cardiac output directly contribute to hypotension when the blood pressure is not adequately compensated by increases in systemic vascular resistance, heart rate, stroke volume or contractility. This commonly occurs in advanced heart failure.

Volume Depletion

Intravascular volume determines venous return, cardiac preload, and cardiac output. Volume depletion reduces venous return, stroke volume, and cardiac output, causing **hypovolemic shock**. Hemorrhage is the most direct cause, with blood loss >30–40% rapidly inducing shock; common sources include trauma, surgery, and GI bleeding. Non-hemorrhagic losses—vomiting, diarrhea, sweating, burns—and renal losses from diuretics, diabetes insipidus, or osmotic diuresis also contribute. **Third-spacing**, where fluids shift to interstitial spaces, in conditions like pancreatitis, cirrhosis, nephrotic syndrome, and obstruction. Relative hypovolemia occurs when vascular capacity expands, as in pregnancy.

CLINICAL CORRELATION

Inadequate fluid intake, often seen in elderly patients or those with impaired consciousness or access to fluids, can exacerbate volume depletion from other causes or independently lead to hypotension if prolonged.

What Are the R onsequences of Hypotension? (LO#3)

Hypotension can range from a benign finding in some individuals to a life-threatening condition in others. The risk of developing hypotension and the severity of its consequences depend on various factors, including patient characteristics, comorbidities, and the underlying cause.

Risk Factor Profiles

Age significantly influences the risk of hypotension, with elderly individuals being particularly vulnerable due to age-related changes in cardiovascular physiology. Additionally, older adults often have comorbidities and take multiple medications that can predispose them to hypotension. The prevalence of orthostatic hypotension, for example, increases from approximately 5% in middle-aged adults to 30% or higher in those over 75 years.

Medication use represents one of the most modifiable risk factors for hypotension. Several drug classes can lower blood pressure as either a therapeutic effect or an adverse reaction.

Common medications associated with hypotension include:

- **Antihypertensives:** ACE inhibitors, angiotensin receptor blockers, calcium channel blockers, diuretics, and alpha-1 blockers (common drugs causing hypotension)
- **Cardiovascular drugs:** Nitrates, amiodarone, and phosphodiesterase inhibitors (common drugs causing hypotension)
- **CNS-acting:** Antipsychotics (particularly phenothiazines), tricyclic antidepressants, and MAO inhibitors
- **Anesthetics and** oids, and epidural/spinal anesthesia
- **Others:** Dopamine agonists (for Parkinson's disease)

Which class of medications can cause hypotension by acting on the CNS?

Tissue Perfusion Complications

The primary consequence of hypotension is inadequate tissue perfusion affecting multiple organ systems. The brain is especially susceptible due to its high metabolic demand and limited energy reserves. Cerebral autoregulation normally maintains blood flow, but when mean arterial pressure falls below approximately 60 mmHg, perfusion becomes pressure-dependent. Acute cerebral hypoperfusion may cause dizziness, lightheadedness, visual changes, confusion, syncope, or seizures. Sustained or severe reductions in cerebral blood flow can result in ischemic stroke or global hypoxic-ischemic brain injury.

Reduced coronary perfusion in coronary artery disease can trigger ischemia, leading to angina, arrhythmias, or infarction, and worsening cardiac output in a vicious cycle.

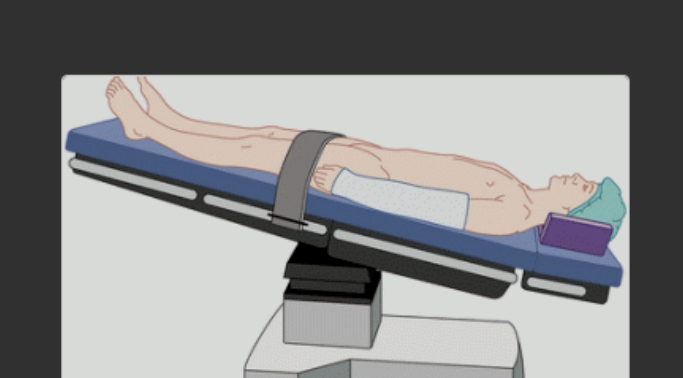
Renal perfusion is normally autoregulated but becomes pressure-dependent when mean arterial pressure falls below approximately 80 mmHg, triggering activation of the renin–angiotensin–aldosterone system (RAAS) to support blood pressure. While initially compensatory, sustained renal hypoperfusion reduces renal blood flow and glomerular filtration rate, potentially leading to acute kidney injury with elevated creatinine, oliguria, and electrolyte disturbances.

What Are the Treatment Strategies for Hypotension Including Nonpharmacological Approaches? (LO#4)

Management of hypotension involves a stepwise approach using supportive measures and nonpharmacological strategies to improve venous return, volume, vascular tone, and cardiac function.

Immediate Supportive Measures

The initial management of acute hypotension focuses on stabilizing the patient and preventing end-organ damage while diagnostic evaluation proceeds. Positioning is a simple yet effective intervention for hypotensive patients. The Trendelenburg position (supine with legs elevated) or modified Trendelenburg (supine with only legs elevated) increases venous return to the heart by utilizing gravity to shift blood from the lower extremities to the central circulation. This can temporarily improve cardiac preload and cardiac output, raising blood



Trendelenburg position

For patients with orthostatic hypotension, slow, gradual position changes from lying to sitting to standing can minimize blood pressure drops.

Lifestyle Modifications

Hydration is essential for preventing and managing hypotension, particularly in volume depletion or autonomic dysfunction. Patients should increase fluid intake to **2–3 liters daily**, unless contraindicated by heart failure or kidney disease. **Dietary sodium supplementation** benefits those with neurogenic orthostatic or chronic hypotension without contraindications such as hypertension or renal impairment. Increasing sodium intake to **6–10 grams per day** (about 1–2 teaspoons of salt) expands plasma volume and improves orthostatic tolerance. These measures help maintain intravascular volume and reduce hypotensive symptoms.

Physical countermaneuvers can help maintain blood pressure during position changes or prolonged standing. These techniques increase venous return and peripheral vascular r and include:

- Leg crossing and tensing of leg muscles while standing
- Squatting or bending forward
- Toe raises and heel-to-toe rocking
- Contraction of abdominal and buttock muscles
- Avoiding prolonged standing and using a counter or chair for support when necessary

Compression garments, such as waist-high compression stockings (30-40 mmHg pressure), abdominal binders, or full-body compression suits, can reduce venous pooling in the lower extremities and splanchnic circulation, thereby improving venous return and cardiac output. These garments are particularly useful for patients with neurogenic orthostatic hypotension or autonomic dysfunction.

What are key nonpharmacological approaches for managing orthostatic hypotension?

What Pharmacological Treatment Options Are Used to Manage Hypotension?

Pharmacological interventions for hypotension are typically reserved for cases where the underlying cause cannot be fully corrected or when nonpharmacological measures provide insufficient relief.

Volume Expanders

Volume expansion is a cornerstone of managing hypotension due to absolute or relative hypovolemia. Intravenous fluid therapy aims to increase intravascular volume, improve venous return and cardiac preload, and ultimately enhance cardiac output.

Crystalloid solutions (*Normal saline 0.9%, and Lactated Ringer's*) are the most commonly used intravenous fluids for volume expansion. These solutions contain water and electrolytes but no colloids or large molecular weight substances. (More information regarding these agents will be provided in the Renal organ system course)

Vasopressors and Inotropic Agents

Vasopressors and inotropic agents are medications that increase blood pressure by enhancing vascular tone, cardiac contractility, or both. These agents are typically used when fluid resuscitation alone is insufficient to restore adequate tissue perfusion. They act primarily through adrenergic or dopaminergic receptor stimulation, or through other mechanisms that influence vascular smooth muscle or cardiac function.

Agents used for H

For chronic orthostatic hypotension, particularly neurogenic orthostatic hypotension, several medications may be considered when nonpharmacological measures are insufficient.

Fludrocortisone acts as a synthetic mineralocorticoid that binds to aldosterone receptors in the distal renal tubules, promoting sodium and water reabsorption. This increases intravascular volume and blood pressure, helping to counteract hypotension. Additionally, it enhances potassium and hydrogen ion excretion, maintaining electrolyte balance.

Midodrine, an α1-adrenergic agonist, increases peripheral vascular resistance. Droxidopa, a synthetic precursor of norepinephrine, increases blood pressure by enhancing peripheral vascular tone.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to EL, what could be causing her hypotension? What are the likely risks for a patient like her? EL experienced orthostatic hypotension likely exacerbated by dehydration and antihypertensive medications. Her combination of hydrochlorothiazide (which can cause volume depletion) and lisinopril (which affects vascular tone) created a perfect storm for hypotension, especially when combined with possible inadequate fluid intake.

Her risk for falls, syncope, and possible organ hypoperfusion highlights the need for targeted interventions discussed in this brick. Management would likely include medication adjustment, improved hydration, and possibly compression garments to prevent recurrent episodes and their potentially serious consequences.

Summary

- Hypotension is defined as systolic blood pressure below 90 mmHg or mean arterial pressure less than 65 mmHg, leading to compromised organ perfusion and oxygen delivery.
- The pathophysiology of hypotension involves impaired vascular tone, reduced cardiac output, and volume depletion, each contributing to decreased blood pressure.
- Risk factors for hypotension include advanced age, cardiovascular and neurological comorbidities, medication use, and acute illnesses, with potential consequences such as falls, syncope, and organ hypoperfusion.
- Nonpharmacological management of hypotension includes supportive measures like positioning and oxygenation, as well as lifestyle modifications such as hydration, increased dietary sodium, and use of compression garments.
- Pharmacological treatments for hypotension involve volume expanders, vasopressors, and inotropic agents, chosen based on the underlying cause and patient-specific factors.
- Orthostatic hypotension is characterized by a significant drop in blood pressure upon standing, and management includes lifestyle changes and, if necessary, medications like fludrocortisone and midodrine.

Review Questions

1. How is hypotension generally defined in clinical practice?

A. ● SBP <90 mmHg or MAP <65 mmHg
B. ○ SBP >140 mmHg or MAP >90 mmHg
C. ○ Fall in SBP >20 mmHg standing
D. ○ Drop in BP after meals
E. ○ Heart rate below 60 bpm

Hide Explanation

The correct answer is (A). Hypotension is typically defined as a systolic blood pressure below 90 mmHg or a mean arterial pressure less than 65 mmHg. Option (B) describes hypertension. Options (C) and (D) relate to specific subsets of hypotension but do not define it. (E) is unrelated to hypotension thresholds.

2. Which mechanism most commonly causes hypotension in systemic inflammatory response syndrome (SIRS)?

A. ○ Decreased stroke volume
B. ○ Impaired cardiac conduction
C. ● Decreased systemic vascular resistance
D. ○ Reduced intravascular volume
E. ○ Pulmonary vascular obstruction

Hide Explanation

The correct answer is (C). Septic shock involves decreased systemic vascular resistance due to vasodilation from inflammatory mediators. (A) Decreased stroke volume typically presents with cool extremities in cardiogenic shock. (B) Impaired conduction causes bradycardia, not tachycardia seen here. (D) Reduced volume causes hypovolemic shock, often with cool skin. (E) Pulmonary obstruction causes obstructive shock with respiratory signs.

3. An 82-year-old woman with Parkinson's disease experiences dizziness when standing. Which complication is she at highest risk for?

A. ○ Myocardial infarction
B. ○ Mesenteric ischemia
C. ○ Acute kidney injury
D. ● Falls

Hide Explanation

The correct answer is Falls (D). Elderly patients with Parkinson's disease commonly develop orthostatic h ng that significantly increases fall risk. Myocardial infarction (A) is possible with severe hypotension but less directly linked to orthostatic changes. Mesenteric ischemia (D) typically occurs with profound, sustained hypotension rather than orthostatic changes. Acute kidney injury (C) generally requires more prolonged hypoperfusion than brief orthostatic episodes.

4. Which medication increases intravascular volume by promoting sodium reabsorption?

A. ● Fludrocortisone
B. ○ Midodrine
C. ○ Pyridostigmine
D. ○ Droxidopa

Hide Explanation

The correct answer is Fludrocortisone (A). Fludrocortisone acts as a synthetic mineralocorticoid that binds to aldosterone receptors, promoting sodium and water reabsorption. Midodrine (B) is an alpha-1 agonist that increases vascular tone. Pyridostigmine (C) is an acetylcholinesterase inhibitor, not directly affecting sodium reabsorption. Droxidopa (D) is a norepinephrine precursor.

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Define hypotension and identify its primary hemodynamic effects on organ perfusion and tissue oxygenation.~~
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Objective Confidence: 4/4